

Online Assignment Submission

UNIVERSITY OF
BIRMINGHAM

School of Government

Student ID Number: 3059714

Programme: MSc AI and Sustainable Development

Module: AI for Global Challenges

Assignment Title: Assignment One - 1000 words essay worth 75% of overall mark

Date and Time of Submission: 30th March 2026 at 13:01

Actual Word Count: 1098 (including Reference)

Extension: N **Extension Due Date:**

I do wish my assignment to be considered for including as an exemplar in the **School Bank of Assessed Work**. * *delete as appropriate*

Please ensure that you complete and attach this template to the front of all work that is submitted.

It is a requirement to complete the first section, and to include this completed template as the first page of every assignment that is submitted for marking (your School will advise on exceptions).

Declaration

By submitting this work, you confirm that it is your own original work, that it reflects your own thinking, analysis, and understanding, and that it therefore complies with the University's Code of Practice on Academic Integrity.

This includes adhering to university guidance on the [ethical use of generative AI](#).

Please tick the box below to confirm this.

[x] (ticked) I confirm that this is my own original work. I have not included any content from generative AI tools in this assignment, except in ways explicitly permitted in the assignment instructions.

CONTINUED BELOW

Remember to use feedback from previous formative and summative assignments towards this assignment. You can discuss previous feedback with your module or seminar convenors during office hours before submitting. If you wish to receive targeted feedback that will support your learning, please use this second section.

Section One: Reflecting on the feedback that I have received on previous assessments, the following issues/topics have been identified as areas for improvement: (add 3 bullet points). *NB – for first year students/PGTs in the first term, this refers to assessments in your previous institution*

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Section Two: In this assignment, I have attempted to act on previous feedback in the following ways (3 bullet points)

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Section Three: Feedback on the following aspects of this assignment (i.e. content/style/approach) would be particularly helpful to me: (3 bullet points)

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Report: Replication and Critical Assessment

Introduction

This report replicates and critically appraises Celik et al. (2025), a study that uses machine learning on the 2025 SDG Index to group countries by sustainability performance. The topic is relevant because sustainability indicators are high-dimensional, making clustering a natural tool for identifying policy-relevant patterns.

My main finding is that the paper's central five-cluster result is only partly reproducible from the released repository. The clustering pattern can be recovered, but only after reconstructing missing implementation details. The key issue is that the paper's preferred $k=5$ depends on an undocumented K-Means setting ($n_init=1$); under standard settings, $k=4$ or $k=6$ are preferred, making the five-cluster result unstable. The repository does not provide a clean one-command pipeline, some results are hard-coded rather than computed, and the Random Forest results still do not fully match the paper. The study asks an important question, but its release standard is weak.

Summary of the Original Paper

The paper studies country-level sustainability differences using the 2025 SDG Index. It uses cross-sectional data and focuses on SDG score, international spillover score, regional score, population, and progress on headline SDG indicators. The aim is to identify statistically meaningful groups of countries with similar sustainability profiles.

The main method is K-Means clustering. The number of clusters is chosen using the Elbow method and silhouette score, with the paper selecting five clusters. PCA is then used to

visualise the clusters, and average feature profiles are used to interpret them. The paper also reports ANOVA and MANOVA results as evidence that the clusters differ significantly.

The authors also train several classifiers, including Random Forest, SVM, Decision Tree, XGBoost, ANN, and Logistic Regression, to predict the cluster labels. They report high predictive performance and use ROC curves and feature importance to argue that the groupings are stable and policy-relevant.

Summary of My Replication

I worked only from the materials in the released repository: a minimal README, one CSV file, and one large exported notebook script (Codes.py). The README gives no instructions or package versions, and the main script is a notebook dump with repeated cells and state-dependent execution. It does not run cleanly as a standalone script due to syntax errors and missing variable dependencies.

I therefore rebuilt the workflow step by step. First, I checked the effective sample. The CSV contains 167 countries, but the clustering code drops rows with missing values in the five numerical features, leaving 143 observations. This matters because the paper reports a larger sample (166), while the effective sample is 143.

The main technical focus of my replication was Figure 5 and the choice of five clusters. In the released code, K-Means is run with *random_state=42*, but important initialization details, especially *n_init*, are not specified. *n_init* is the number of times K-Means runs with different random starting points, keeping the best clustering result (lowest inertia). That omission matters because WCSS and silhouette values are sensitive to K-Means initialization and to *scikit-learn* version. After aligning preprocessing and testing initialization settings, it

becomes clear that $n_init=1$ reproduces the paper's $k=5$ choice, but $n_init=5$ points to $k=6$ and $n_init=10$ points to $k=4$. Thus, the main clustering result depends on a hidden hyperparameter and is weaker than nearby alternatives.

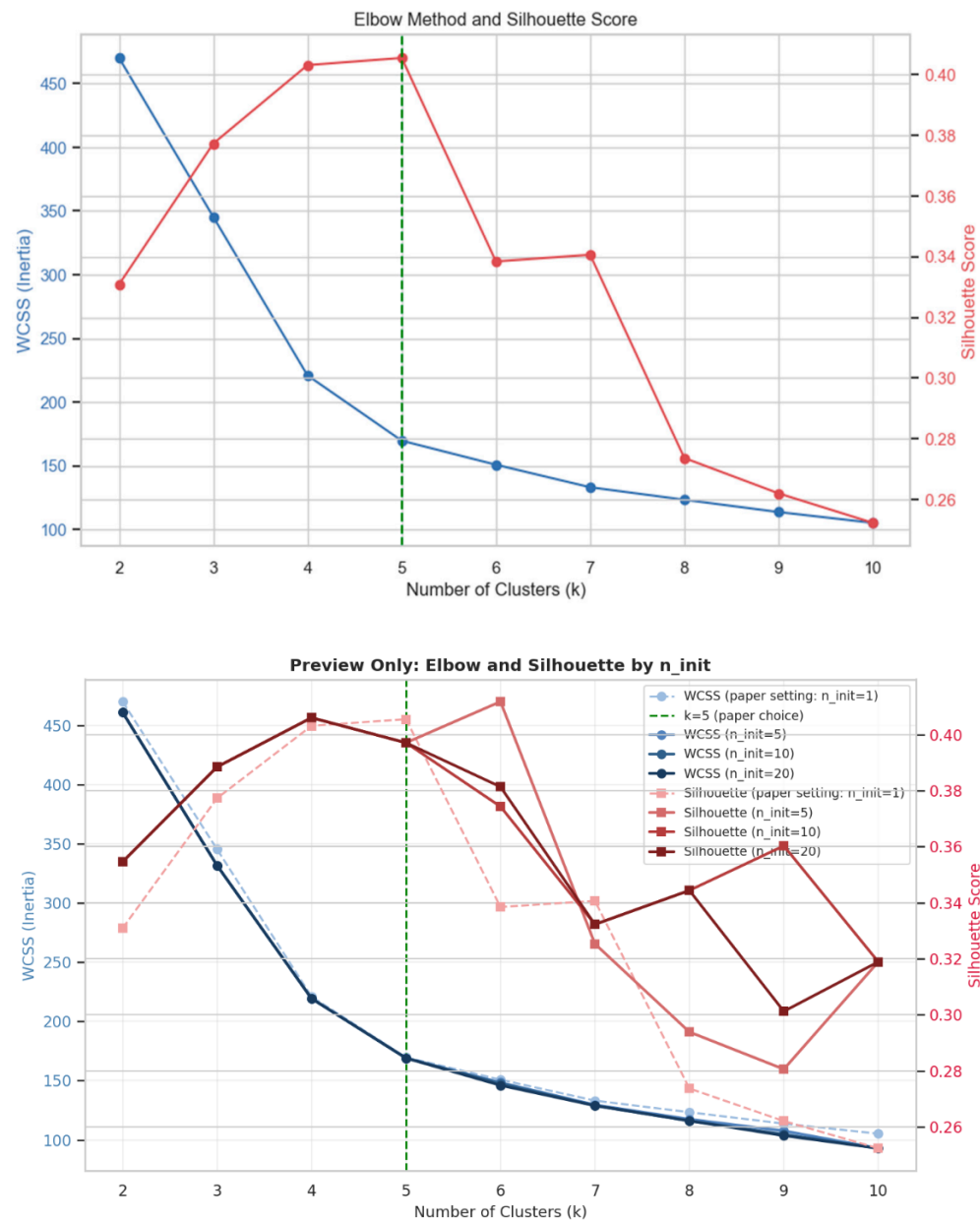


Figure. Original (Celik et al., 2025) vs different n_init

I then reviewed the validation sections. The visual cluster profiling can be reconstructed from the code, but the ANOVA and MANOVA tables cannot be independently replicated

because the script prints manually hard-coded values instead of calculating them. The classification section is also hard to audit because the repository contains several overlapping model-evaluation blocks with slightly different setups. Most of the classification story is similar, but the Random Forest results still do not fully match the paper.

Critical Assessment of the Research Design

The paper uses a cross-sectional, observational design based on secondary country-level data. That is appropriate for descriptive pattern detection, and K-Means plus PCA is a plausible toolkit for summarising sustainability indicators. The issue is not the design itself, but the weak transparency of its implementation.

At the reproducibility level, the released materials do not meet the standard of a clean computational study. There is no meaningful README, no dependency file, no pinned software environment, and no clear instruction for reproducing tables or figures. This is most evident in Figure 5, where the omitted *n_init* hyperparameter determines the result. In my tests, *n_init*=1 yields the paper's five-cluster conclusion, whereas *n_init*=10 favors four clusters and *n_init*=5 favors six.

At the implementation level, there are mismatches between the written methods and the code. The paper describes min-max normalization and inclusion of an encoded region variable. The clustering code does neither. Instead, it uses five numeric variables only and applies *StandardScaler()*. Min-max scaling appears later only for visualizing cluster averages. A reader following the paper text alone would rebuild the wrong pipeline.

Sample reporting is also inconsistent. The paper refers to more countries than are actually used in the clustering stage. The released CSV has 167 rows, but the effective sample is 143 after dropping missing progress values, and the published cluster sizes also sum to 143.

At the validation stage, the ANOVA and MANOVA tables are created from manually typed dictionaries rather than from computed outputs, and the comments describe them as example values. This makes the statistical validation unverifiable.

The classification section is also weaker than it first appears. The repository contains overlapping evaluation blocks, and the Random Forest setup is not consistent across them. One section uses a standalone Random Forest on an unstratified split for feature importance, while later sections use scaled inputs and stratified sampling in a multi-model comparison. This makes it unclear which specification corresponds to the published results and helps explain why the Random Forest row still does not match cleanly.

There is also a broader design issue. The supervised models are trained to predict cluster labels that were themselves created from the same input variables. High classifier accuracy therefore shows internal separability, not strong external validation of real sustainability groupings. Overall, the study is more convincing as exploratory than as a fully reproducible AI application. The most important replication lesson is that the published five-cluster result is not stable unless one reconstructs the undocumented $n_{init}=1$ choice used in K-Means.

The release would be stronger with a clean script, explicit hyperparameters, pinned versions, consistent sample reporting, and computed tables.

Reference

Çelik, S., Öztürk, Ö. F., Akkucuk, U., & Şaşmaz, M. Ü. (2025). Global sustainability performance and regional disparities: A machine learning approach based on the 2025 SDG Index. Sustainability, 17(16), 7411. <https://doi.org/10.3390/su17167411>

Authors' original repo: <https://github.com/Sadullah4535/Global-Sustainability-Performance->

Student's replication repo: <https://github.com/qmanhbeo/AI4GC-1-submission-Leo>